# AKARSH NAVHULE

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## PROFESSIONAL SUMMARY

Full Stack AI Engineer with 1.5+ years building production AI systems — RAG pipelines, LLM agents, diffusion model workflows, and GCP-deployed applications. Experienced shipping AI products end-to-end from model integration to frontend.

## TECHNICAL SKILLS

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|---|---|
| **Languages** | Python, JavaScript |
| **Frontend** | React.js, Next.js, Tailwind CSS |
| **Backend** | FastAPI, Node.js, REST API Design |
| **AI / ML** | RAG, vector search, embeddings, fine-tuning (LoRA), agentic workflows, multimodal — ChromaDB, LLM APIs (Gemini, OpenAI), ComfyUI, n8n |
| **Databases** | MongoDB, PostgreSQL, ChromaDB |
| **Cloud & DevOps** | GCP (Compute Engine, Cloud Storage), Nginx, Shell scripts |

## WORK EXPERIENCE

**Full Stack Developer** · **L&T Finance** — Bengaluru, India — Jul 2024 – Oct 2025

- Built a high-throughput FastAPI service using AWS Rekognition for facial-recognition attendance, handling 500+ concurrent API calls/second with async batching and multi-threading.
- Developed a React.js analytics dashboard surfacing real-time attendance states; integrated MongoDB for persistent event logging and audit trails.
- Used ChatGPT for context-aware development; estimated 2x faster delivery compared to prior sprint velocity.

**Freelance Full Stack Developer** · **Self-Employed** — Bengaluru, India — Nov 2025 – Mar 2026

- Built **Malnadrealty.com** ↗ — a NoBroker-style property marketplace; sole engineer for end-to-end GCP deployment (Compute Engine, Nginx, MongoDB, PostgreSQL), OTP login, WhatsApp notifications, and headless WordPress CMS across 150+ REST APIs.

## PROJECTS

### 01 AI Digital Twin — Face, Voice & Video Synthesis ↗ GitHub ↗ Portfolio

*ComfyUI · FLUX LoRA · Chatterbox (voice cloning) · Infinite Talk · Hugging Face · Jarvis Labs GPU · ngrok*

- Fine-tuned a FLUX LoRA on 15 personal photos with custom trigger token "Akarsh"; integrated Chatterbox voice clone into an Infinite Talk ComfyUI workflow for end-to-end lip-synced video generation from a single image + text prompt; **~₹10–15 per 40s video** on Jarvis Labs GPU.
- JarvisLabs shipped an outdated ComfyUI template — manually installed the latest version via Jupyter, then exposed the server publicly using ngrok since it ran on a remote host with no direct URL.

### 02 AI Automation Agents (n8n) ↗ GitHub

*n8n · OpenAI GPT-4.1 · Gmail API · Razorpay · Google Calendar · Sora2 · Google Sheets*

- **WhatsApp Clinic Agent:** n8n webhook trigger → GPT-4.1 intent classification → Razorpay payment confirmation as conditional gate → Google Calendar booking; fully async, zero human input.
- **AI UGC Video Generator:** product photo-to-UGC-style video pipeline using Sora2 + GPT-4.1 mini at **$0.15/video**.
- **Auto Email Labeling Agent:** reads Gmail, classifies and labels emails, auto-drafts replies using GPT-4.1 nano.

### 03 Sahayak — AI Student Assistant (Hackathon) ↗ Backend ↗ Frontend ↗ Demo

*FastAPI · Next.js · ChromaDB · Google Gemini Flash 2.5 · Google Vision API · GCP VM*

- Built a full-stack RAG application that ingests textbooks via OCR, retrieves context through ChromaDB vector search, and uses Gemini Flash 2.5 to generate question papers and evaluate answer sheets.

### 04 Claude Code Agent Projects

*Claude Code · Remotion · YouTube Data API · Google Sheets API · LLM APIs*

- **YouTube Niche Analyzer:** agent pipeline that fetches video metadata, runs LLM analysis on engagement patterns, and auto-generates structured Excel + PPT reports delivered via email; configurable fetch count, reports in 5–10s. ↗ GitHub
- **Motion Graphics Generator:** prompt-driven pipeline using Claude Code + Remotion, rendering to MP4. ↗ GitHub
  **Google Sheet AI Chat:** LLM-powered natural language Q&A over Sheets data. ↗ GitHub

## EDUCATION

**B.E. — Computer Science & Engineering**
**Nitte Meenakshi Institute of Technology, Bengaluru**
CGPA: 8.16 / 10.0 · Jul 2020 – Jun 2024

## CERTIFICATIONS

**100xEngineers GenAI Program** (In Progress)
Diffusion Models · LLM Pipelines · AI Agents · RAG